

THE SOC THAT SHOWS ITS WORKING

A three-agent pipeline detects, explains and contains attacks on critical infrastructure — and never states a fact it cannot cite

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Every year the Indian Computer Emergency Response Team logs more than a million and a half cybersecurity incidents — 1.59 million of them in 2023 alone, a tally that has only climbed since. AIIMS Delhi in 2022. CBSE, twice, in 2024 and again this year. India's public institutions keep getting hit, and more than seven in ten government entities are still running IT infrastructure that reached end-of-life years ago.

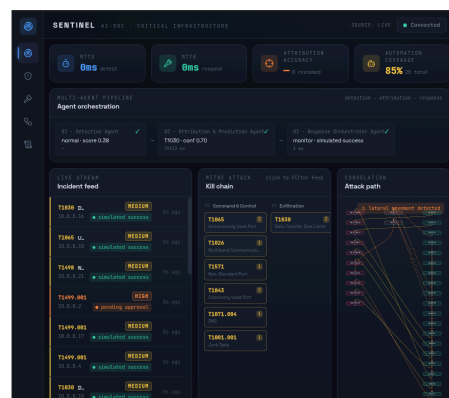
The uncomfortable truth, investigators on this project say, is that the failure was never a shortage of alerts. It is *detection speed*. Signature-based tools cannot see an advanced persistent threat that never matches a known signature and deliberately moves low-and-slow through a network — probing, waiting, moving laterally, waiting again. What has been missing, until now, is a behavioral intelligence layer: something that notices a host is no longer behaving like itself, correlates the weak signals into a kill chain, and acts fast enough that the response happens in hours, not the weeks a

manual security operations centre typically needs.

That is precisely the gap this project, **AI-SOC**, was built to close — not as a slide deck promising a future build, but as a working pipeline, wired end-to-end, running today.

Three coordinating agents do the work. A Detection Agent, an unsupervised Isolation Forest trained only on normal network traffic, flags deviation without ever being shown a labelled attack. An Attribution & Prediction Agent, a retrieval-grounded language model, maps that deviation to a cited MITRE ATT&CK technique and a CVE or CERT-In reference — and is built, deliberately, so that it cannot invent a technique ID it cannot source. A Response Orchestrator then applies a fixed policy table to decide what happens next, holding every high-impact action at a human-approval gate. Every step, without exception, is written into a tamper-evident, hash-chained audit log.

CONTINUED ON PAGE 2 — “THREE AGENTS. NOT ONE DASHBOARD.”



THE DASHBOARD, LIVE — the incident feed streaming in real time over server-sent events; the same screen a duty analyst would watch. *Photo: AI-SOC engineering team.*

KEY NUMBERS

0.867

ROC-AUC, threshold-free — Isolation Forest on UNSW-NB15

0.749 / 0.820

Recall at the balanced (10.0% FPR) and high-recall (20.3% FPR) operating points

0.981

Precision at the balanced operating point

3

Coordinating agents — detection, attribution, response — feeding one live dashboard

100%

Of automated actions hash-chained and gated for human approval above a blast-radius threshold

Public infrastructure keeps getting hit

CONTINUED FROM PAGE 1

The scale is not in dispute. CERT-In alone handled **1.59 million** cybersecurity incidents in 2023, a volume that has continued to surge every year since records were centralised. Public institutions have borne a disproportionate share of the damage: AIIMS Delhi's systems were knocked offline for weeks in 2022; CBSE's examination infrastructure was hit in both 2024 and 2026. These are not obscure targets — they are hospitals and examination boards that hundreds of millions of citizens depend on.

Behind the headlines sits a more structural problem. Over **70%** of government entities are still running critical workloads on end-

of-life IT infrastructure — systems that no longer receive vendor security patches, sitting exposed on networks that are, by design, supposed to be among the most defended in the country.

“The failure was never a lack of alerts — it's detection speed.”

— PROJECT DESIGN NOTE, AI-SOC ENGINEERING TEAM

Signature-based tools remain the default defence across most of this infrastructure, and they share one structural weakness: they can only catch what they have already been told to look for. An advanced

persistent threat that never repeats a known signature, and that deliberately moves low-and-slow to stay under alerting thresholds, simply does not trip them.

What is missing, this project's engineering notes argue, is a **behavioral intelligence layer** — a system that learns what “normal” looks like for a given host or network segment, notices when that baseline is violated, correlates the weak signals it finds across a full kill chain, and is fast enough that the resulting response happens in hours rather than the weeks a conventional security operations centre needs to triage, attribute and act.

HOW IT WORKS

Three agents. Not one dashboard.

Detection, attribution and response are framed and wired as genuinely coordinating agents, feeding a single live dashboard

01 DETECTION	02 ATTRIBUTION	03 RESPONSE	04 DASHBOARD
Isolation Forest Unsupervised, no signatures anywhere. Learns “normal,” flags what isn't. ----- OUTPUT → ANOMALY_EVENT	RAG + MITRE Over ATT&CK, CVE, CERT-In — cited, never invented. Maps anomaly to technique + next stage. ----- OUTPUT → ENRICHED_INCIDENT	Orchestrator Policy-driven, human-approval gate above a blast-radius threshold. Hash-chained audit, SSE live stream. ----- OUTPUT → CONTAINMENT_ACTION	Live Console Live feed, MITRE map, attack-path graph, approvals, audit trail, metrics — one screen, the whole pipeline.

No signatures. The detection model learns only what normal traffic looks like; it flags deviation, not a known pattern — the entire point when an attacker is specifically evading signature matching.

Retrieve-then-cite, always. The attribution agent never states a technique ID it cannot ground in a retrieved source. Where retrieval comes back thin, the stated confidence drops honestly instead of guessing.

Policy decides, not the model. A single authoritative policy table maps anomaly score and severity to an action; the agent's suggestion is advisory only, and high-impact actions wait for a human above a blast-radius threshold.

Agent dossiers: who does what

CONTINUED FROM PAGE 2 | REPORTING FROM ENGINE / , INTEL / , ORCHESTRATOR /

01 • The Detection Agent

Housed in **engine/**, the detection agent is an unsupervised Isolation Forest — 400 trees, fit on 69,750 rows of pooled normal UNSW-NB15 traffic, never once shown a labelled attack in training. At inference it scores live flows and reports the specific human-readable features that made a flow look abnormal, not a bare score.

Its output, the `anomaly_event` contract, carries the score and `top_features` in plain language, plus the raw vector the downstream agent reasons over; source and destination IP double as the attack-path graph's edges.

02 • The Attribution & Prediction Agent

Where the detector notices, this agent explains. Living in **intel/**, it is a tool-calling agent (Claude, Groq fallback) on a strict retrieve-then-cite discipline: it queries a Chroma index built from MITRE ATT&CK STIX, NVD CVE and CERT-In advisories, and cannot name a technique ID

it cannot ground in a retrieved passage. Thin retrieval drops the stated confidence honestly, rather than guessing — and it predicts the likely next attack stage alongside the current one.

03 • The Response Orchestrator

In **orchestrator/**, the design turns deliberately conservative: `policy.py` alone maps an anomaly's score and severity to an action — the model's suggestion is advisory only. Actions above a blast-radius threshold sit at `pending_approval` until a human approves. Every action, approved or not, is hash-chained (`entry_hash = sha256(prev_hash + entry)`) and re-verified client-side.

04 • The Dashboard

All three agents surface on one screen: live incident feed, MITRE technique grid, attack-path graph, approval queue and audit view — the same proof-of-pipeline a duty analyst would use.

THE SEAM

Three contracts. Nothing drifts.

Three frozen JSON contracts, mirrored as Pydantic models — the seam every pillar imports, so nothing changes silently out of sync

ANOMALY EVENT	ENRICHED INCIDENT	CONTAINMENT ACTION
<code>event_id</code> — e.g. <code>"evt_0001"</code>	<code>attack_technique</code> — cited MITRE ATT&CK id	<code>action</code> — <code>isolate_host</code> / <code>block_ip</code> / <code>revoke_credential</code> / <code>snapshot_vm</code> / <code>monitor</code>
<code>anomaly_score</code> — threshold-free	<code>confidence</code> / <code>severity</code> — low retrieval → lower stated confidence	<code>status</code> — <code>pending_approval</code> → <code>approved</code> → <code>simulated_success</code>
<code>top_features</code> — human-readable, not a column index	<code>cve_refs</code> / <code>certin_refs</code> — grounded citations	<code>requires_human_approval</code> — true above blast-radius
<code>raw_features</code> — real numbers the agent reasons over	<code>suggested_action</code> — advisory only	<code>audit_log_id</code> — hash-chained entry reference
<code>src_ip</code> + <code>dst_ip</code> — graph edges	<code>predicted_next</code> — likely next attack stage	<code>actor</code> — system or approving human

A defensible number, not perfection

Unsupervised, fit on normal traffic only, evaluated on held-out normal plus every attack class in UNSW-NB15

ROC-AUC	RECALL @ BALANCED	RECALL @ HIGH-RECALL	PRECISION @ BALANCED
0.867 threshold-free	0.749 at 10.0% FPR	0.820 at 20.3% FPR	0.981 optimum threshold

THE FULL SWEEP

TRAIN-NORMAL PERCENTILE	THRESHOLD	RECALL	FPR	PRECISION	F1
70	0.4097	0.852	0.305	0.952	0.899
75	0.4154	0.840	0.254	0.959	0.895
80	0.4223	0.820	0.203	0.966	0.887
85	0.4316	0.787	0.152	0.973	0.870
90 (default)	0.4456	0.749	0.100	0.981	0.849
95	0.4747	0.597	0.049	0.988	0.745
98	0.5203	0.442	0.020	0.994	0.612

Source: engine/RESULTS.md. Per-class detail: the model catches loud attacks (Generic ≈0.99, DoS ≈0.89, Exploits ≈0.80 recall) and misses low-and-slow classes (Reconnaissance, Shellcode, Fuzzers) that are near-benign in flow statistics — a known limit of unsupervised flow detection, which is why detection is stage one of three, not the whole system.

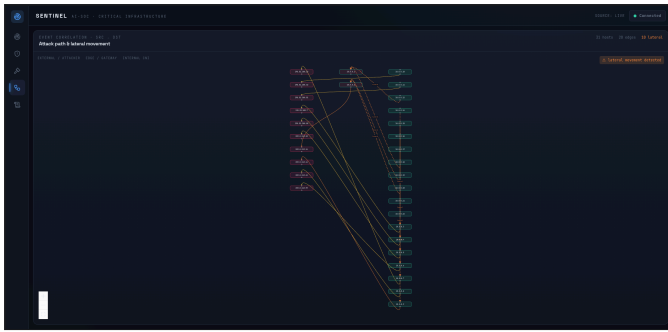
SCORECARD

Scored on the judges' own terms

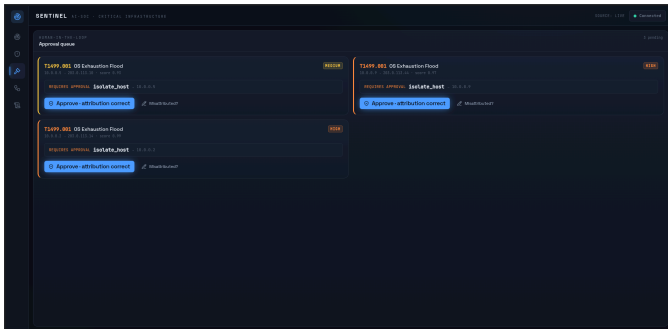
THE BRIEF ASKS FOR	THIS PROJECT'S ANSWER
Anomaly detection rate / FPR on benchmark data	ROC-AUC 0.867; full recall/FPR table above
APT attribution accuracy at ATT&CK technique level	Labelled evaluation set; retrieve-then-cite grounding, zero invented IDs
Incident-response automation coverage	autonomous_steps / total_steps, surfaced live on the dashboard
MTTD / MTTR vs. a baseline SOC	Deviation → decision → simulated containment in seconds, not the weeks a manual SOC needs
Full auditability of every automated action	Hash-chained JSONL audit log, independently re-verified client-side

Not slideware — screenshots of the running system

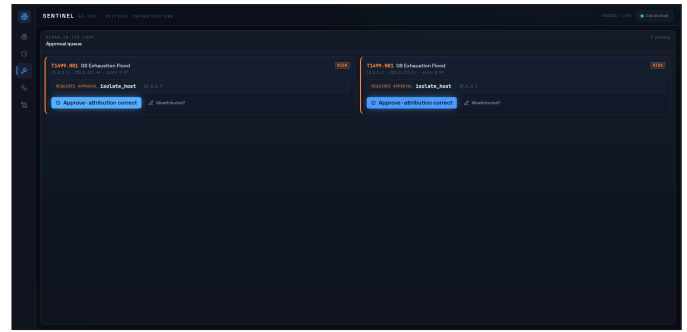
Every image below is a live capture of the actual dashboard, taken against the real backend



ATTACK-PATH GRAPH — events correlated by host into a source-to-destination graph; the orange path marks a detected lateral-movement pattern across 31 hosts and 20 edges.



APPROVAL QUEUE — human-in-the-loop containment requests, each carrying the attributed technique, confidence score and requested action, waiting on a human decision.



AFTER APPROVAL — the same queue once an analyst has approved containment; the action moves to simulated_success and is written to the audit log.

#	EVENT	ACTION	DETAIL	PREV	HASH
1	evt_live_0000	block_ip	T1590.002 (medium, conf 0.70) → block_ip on 203.0.113.87; The anomaly event evt_live_0000.		f4cd47c1...
2	evt_live_0002	block_ip	T1065 (medium, conf 0.70) → block_ip on 203.0.113.65; The detected anomaly evt_live_0002.	f4cd47c1...	4245e6bb...
3	evt_live_0003	block_ip	T1071.004 (medium, conf 0.80) → block_ip on 10.0.0.19; Based on the anomaly event, there...	4245e6bb...	8608eadd...
4	evt_live_0004	isolate_host	T1190 (high, conf 0.80) → isolate_host on 10.0.0.29; The network anomaly event evt_live...	8608eadd...	6cf9a26c...
5	evt_live_0005	block_ip	T1498.001 (high, conf 0.80) → block_ip on 198.51.100.68; The network anomaly event evt-l...	6cf9a26c...	59496e78...
6	evt_live_0001	block_ip	T1043 (medium, conf 0.80) → block_ip on 203.0.113.79; The network anomaly event evt_live...	59496e78...	557e05fc...
7	evt_live_0005	isolate_host	T1590.002 (high, conf 0.95) → isolate_host on 10.0.0.25; The anomaly event evt_live_0005...	557e05fc...	725778ac...
8	evt_live_0006	block_ip	T1071.004 (medium, conf 0.80) → block_ip on 10.0.0.27; An unusual DNS query from source ...	725778ac...	929f01f7...
9	evt_live_0010	block_ip	T1498 (medium, conf 0.70) → block_ip on 198.51.100.46; The network anomaly event evt_liv...	929f01f7...	75d8b4fa...
10	evt_live_0000	block_ip	T1547.015 (medium, conf 0.80) → block_ip on 10.0.0.15; The anomaly event evt_live_0000 i...	75d8b4fa...	f6d4c197...
11	evt_live_0001	block_ip	T1190 (medium, conf 0.70) → block_ip on 10.0.0.27; The anomaly event evt_live_0001 has a...	f6d4c197...	18cb14cc...
12	evt_live_0002	block_ip	T1030 (medium, conf 0.80) → block_ip on 10.0.0.3; The network anomaly event evt_live_000...	18cb14cc...	6f5d6da9...
13	evt_live_0003	block_ip	T1499.003 (medium, conf 0.80) → block_ip on 198.51.100.86; The anomaly event evt_live_00...	6f5d6da9...	3117aa06...
14	evt_live_0000	block_ip	T1568.003 (medium, conf 0.70) → block_ip on 198.51.100.54; The dst_port=53 and high rate...	3117aa06...	3e15ed9f...
15	evt_live_0002	isolate_host	T1498 (high, conf 0.80) → isolate_host on 10.0.0.11; The network anomaly event evt_live...	3e15ed9f...	0667615a...
16	evt_live_0003	block_ip	T1071.001 (medium, conf 0.80) → block_ip on 10.0.0.18; The anomaly event suggests a pote...	0667615a...	e0c23276...
17	evt_live_0005	block_ip	T1590.002 (medium, conf 0.95) → block_ip on 198.51.100.16; The anomaly event suggests un...	e0c23276...	702d01af...
18	evt_live_0006	block_ip	T1043 (medium, conf 0.80) → block_ip on 10.0.0.27; The network anomaly event evt_live_00...	702d01af...	1dadcab1...
19	evt_live_0010	block_ip	UNKNOWN (low, conf 0.00) → block_ip on 203.0.113.19; Attribution failed: LLM provider ca...	1dadcab1...	95d81107...
20	evt_live_0011	block_ip	UNKNOWN (low, conf 0.00) → block_ip on 10.0.0.17; Attribution failed: LLM provider call ...	95d81107...	c4a3df96...
21	evt_live_0016	block_ip	UNKNOWN (low, conf 0.00) → block_ip on 10.0.0.13; Attribution failed: LLM provider call ...	c4a3df96...	ec722e0f...
22	evt_live_0018	block_ip	UNKNOWN (low, conf 0.00) → block_ip on 203.0.113.69; Attribution failed: LLM provider ca...	ec722e0f...	5349ce10...
23	evt_live_0019	block_ip	UNKNOWN (low, conf 0.00) → block_ip on 198.51.100.28; Attribution failed: LLM provider c...	5349ce10...	ee7259e0...
24	evt_live_0020	block_ip	UNKNOWN (low, conf 0.00) → block_ip on 203.0.113.25; Attribution failed: LLM provider ca...	ee7259e0...	c181831c...
25	evt_live_0021	block_ip	UNKNOWN (low, conf 0.00) → block_ip on 198.51.100.44; Attribution failed: LLM provider c...	c181831c...	774de7cb...

TAMPER-EVIDENT AUDIT TRAIL — every containment action hash-chained to the one before it; each row carries its own prev-hash and computed hash.

#	EVENT	ACTION	ACTOR	DETAIL	PREV	HASH
1	evt_live_0000	block_ip	system	T1590.002 (medium, conf 0.70) → block_ip on 203.0.113.87; The anomaly event evt_live_0000.		f4cd47c1...
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CHAIN INTEGRITY, VERIFIED — the audit log's hash chain re-computed and verified client-side on demand, independent of whatever the server claims.

Seven phases, one golden rule

There is always something that runs — one mock replaced at a time, main stays green throughout

PHASE 0	Freeze the seam — contracts + fixtures	DONE
PHASE 1	Walking skeleton — event → decision → approval gate → live feed	DONE
PHASE 2	Real detection swapped in	DONE
PHASE 3	Real attribution agent — the differentiator	DONE
PHASE 4	The winning bolt-ons — multi-agent framing, attack-path graph, metrics, OT nod	DONE
PHASE 5	Dashboard polish	DONE
PHASE 6	Deliverables + pitch — architecture diagram, pitch deck, demo video	IN PROGRESS

STATUS REPORT

**This already runs end-to-end. Today. Not
slideware.**

Detection, attribution, orchestration and dashboard pillars are built and
wired together — not slides promising a future build.

Abhinav Singh & Viscous106 · ET AI Hackathon 2026 · Problem Statement 7

github.com/OfficialAbhinavSingh/et-hackathon-ps7 · Tech stack: FastAPI · Pydantic v2 · scikit-learn · ChromaDB · Claude (Groq fallback) · React 19 + Vite + Tailwind
Prepared as the detailed submission document for ET AI Hackathon 2026, Problem Statement 7 — “AI-Driven Cyber Resilience for Critical National Infrastructure.”